The Capability of AI and ML to recognize AI-Generated Images

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1) Introduction

- The rapid advancement of generative artificial intelligence (GenAI) has led to the creation of highly realistic synthetic images, raising concerns about misinformation, fraud, and security threats.
- This has only worsened as GenAI architectures developed, such as Generative Adversarial Networks (GANs) and Latent Stable Diffusion models, have made it more difficult for the human eye to distinguish between real or fake images.
- However, image classification models, particularly Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs), have shown promise in distinguishing between real and AI-generated images. The objective of this study is to evaluate the performance of these deep learning models against each other to observe which will perform the best. .

The main objectives of the research are:

- The collection and creation of datasets, which will then be used to fine-tune commonly used image classification models, into binary classifiers for AI image detection
- The evaluation of the trained models against each other on the same test set on a variety of metrics, which will be collected in a table below
- The comparison of models trained on generated images of one architecture tested on generated images of another architecture (i.e. A GAN-trained CNN tested on detecting diffusion-generated images), and evaluating the possibility of a universal detector

2) Methodology

- For this research we make use to two image classification architectures to compare: **CNNs** and **ViTs**. For CNN, we chose a ResNet50 model, and for Vision Transformers we chose a 1base 16x16 ViT model.
- For generated images, we make use of the two most popular type of generator architectures, GANs and Diffusion models.
- For training the models we make use of 3 training datasets: A **GAN-based**, **Diffusion-based**, **and Hybrid** dataset. Each dataset comprises of real images and fake images. The main difference between these datasets is generator of the fake images, like with the GANs dataset, having generated images from GANs. The Hybrid dataset is comprised of fake images from all the different used generator architectures.
- Therefore we form a pair, consisting of a model type, and training set to create am AI detection models Since there are 2 model types and 3 training datasets, it results in 6 different unique models that we are going to compare: **GAN-trained CNN, Diffusion-Trained CNN, Hybrid-Trained CNN, GAN-trained ViT, Diffusion-Trained ViT, Hybrid-Trained ViT**.
- These models are then evaluated based on two testing sets: One similar to the Diffusion training set, and one similar to the GAN training one.
- They will be evaluated based on the following metrics:
  - **Accuracy (Acc):** The percentage of 'correct' classification, i.e. real images being flagged as real, and AI-generated being flagged as generated.
  - **Real Accuracy (Real Acc):** The percentage of real images being classified as real images
  - **Ai Accuracy (Ai Acc):** The percentage of AI images being classified as AI images
  - **False Positive (FP):** The percentage of real Images flagged as AI-generated
  - **False Negative (FN):** The percentage of AI-generated images flagged as real

3) Results

	Accuracy	Real Acc	Ai Acc	FP	FN
CNN GAN	98.1	99.4	96.8	0.6	3.2
CNN Diff	61.1	98.8	23.4	1.2	76.6
CNN Hyb	99.8	100	99.6	0	0.4
ViT GAN	89.2	97.4	81	2.6	19
ViT Diff	49.1	97.4	0.8	2.6	99.2
ViT Hyb	90.1	89	91.2	11	8.8

Table of results for GAN Testing Set

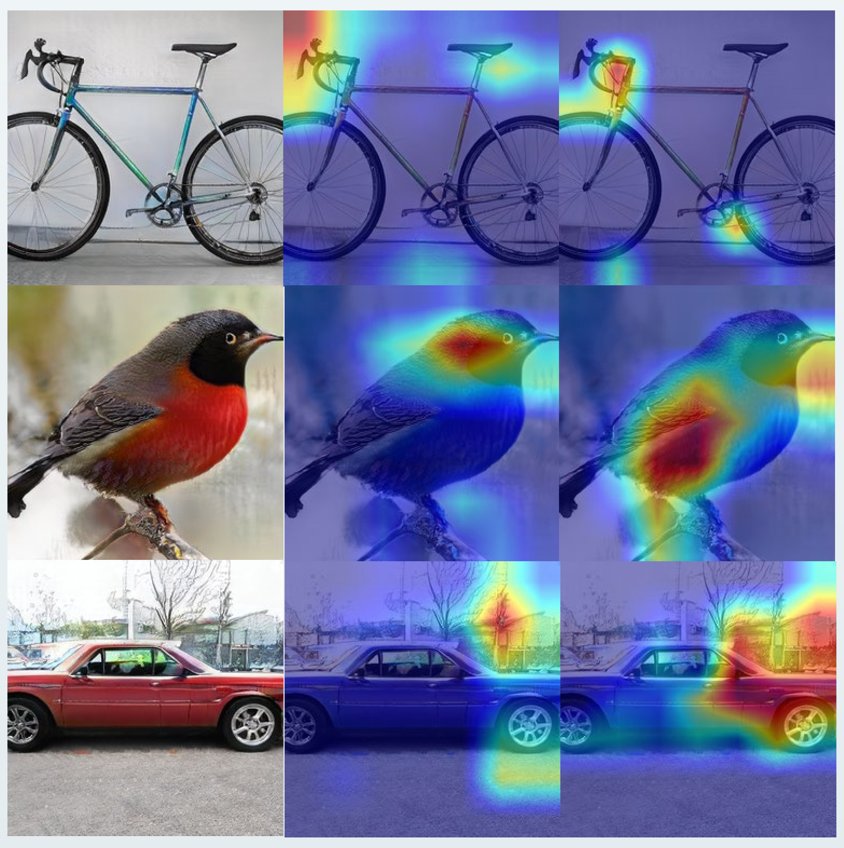
- The results of the GAN-trained models indicate that **GAN images are very easily detected**, with all models achieving at least above 90% accuracy when identifying GAN-generated images.
- Furthermore, for GAN-trained models the **CNN were performing with much higher accuracy than their ViT** counterparts, with nearly 10% differences between them.
- The accuracy of the **Diffusion-trained models is far worse**, indicating that the **artefacts found in diffusion-generated images are different and unapplicable to detecting GAN-generated images**.
- The **Hybrid-trained and GAN-trained model perform similarly**, despite the Diffusion-trained models clearly performing much worse, but the hybrid dataset still maintains greater accuracy, which indicates that the diffusion images used for training did not 'poison' the training of the Hybrid model.
- Overall, all the models were still able to correctly identify real images as real; **the common mistake being the false negatives,** contributing to the inaccuracies of the worse-performing models.

	Accuracy	Real Acc	Ai Acc	FP	FN
CNN GAN	70.7	99.6	41.8	0.4	58.2
CNN Diff	95	99.2	90.8	0.8	9.2
CNN Hyb	93.8	99.6	88	0.4	12
ViT GAN	51.1	98	4.2	2	95.8
ViT Diff	87.8	97.4	78.2	2.6	21.8
ViT Hyb	83.3	89.2	77.4	10.8	22.6

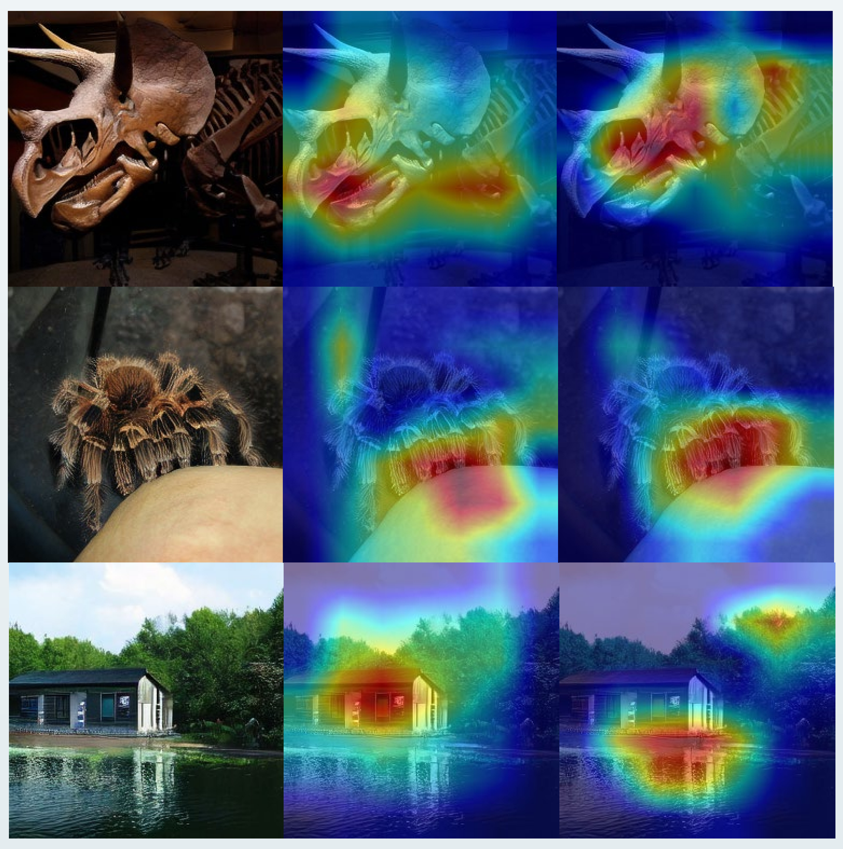
Table of results for DiffusionTesting Set

- The performance on the **Diffusion test datasets shows similar trends to the GAN dataset,** with the Diffusion or Hybrid-trained models performing similarly, while the models trained on a different generator architecture, the GAN-trained models, performed the worst.
- However, the overall accuracy on models trained on the same generator type is worse for Diffusion images compared to GAN images, which could indicate **greater difficulty when identifying Diffusion-generated images**. Other than that, ViTs also perform worse than their CNN counterparts again.
- Additionally, the GAN-trained CNN, though performing much worse than the Diffusion-trained CNN, still achieved 70% accuracy, much better than the 60% achieved by the Diffusion-trained CNN tested on a GAN dataset, which could indicate that artefacts found in GAN-generated images being more universal than Diffusion-generated images.
- Furthermore, once again the inaccuracies can be mostly attributed to False negatives being made, as Real accuracy remains high.

Furthermore, we also decided to utilise a **GradCAM** on the CNN models, to further investigate the difference in the feature considered important between the different CNN Models, by observing the generated heatmaps, from correctly detected fake AI images:



GAN images and their generated heatmaps. (Left column are the original images, middle column is the heatmap generated by GAN-trained CNN, and right column is the heatmap generated by Hybrid-trained CNN.)



Diffusion images and their generated heatmaps. (Left column are the original images, middle column is the heatmap generated by Diffusion-trained CNN, and right column is the heatmap generated by Hybrid-trained CNN.)

- As can be seen from the images above, the heatmaps generated from a Specialty-Trained CNN (A model trained on only one generator architecture, e.g. the GAN or Diffusion trained models) and a Hybrid-trained CNN are significantly different.
- The **general location of the hotspots found in the Specialty-trained can be found in the Hybrid-trained CNN too**, though the specific size and location of these 'shared' hotspots can differ.
- The main difference between the Specialty-trained and Hybrid-trained heatmaps, is the fact that the **Hybrid-trained heatmaps contain hotspot regions that cannot be found in the Specialty-trained ones**.
- Furthermore these hotspots, are generated and are not just small differences, but are actually quite large, even having a similar size to the 'shared' hotspots.
- Though it can be said that the for the **Diffusion images the Hybrid-trained models and the respective Specialty-trained models, have a lesser difference**, with sometimes the hotspots in the Hybrid ones can be found in the Specialty ones, but are less 'hot', or having little difference in general.

4) Conclusion and Future Work

- Overall, it can be seen that **current deep learning image classification models are able to identify and detect AI-generated images**.
- Furthermore, it can also be seen that **different generator types, have different artefacts to be detected**, and that a **detector trained on one type will have difficulties detecting other types of generated images**.
- To overcome this issue, training detectors on **hybrid datasets has shown to maintain similar performances with Specialty-trained models and their respective generator architecture**, but still being able to detect other types of AI images accurately.
- Furthermore, it can be seen that **CNNs perform better than ViTs in AI image detection**.
- Detection of AI-generated images is an ever-evolving issue, with the development and improvement of both image generators and classification models. Hence, a future research study will conduct experiments on a greater number of images, with a specific focus on differing subtypes of each architecture such as the different type of GAN generator, such as CycleGAN or BigGAN.

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